HEALTHCARE CASE STUDY

Dagents for Sepsis

A multi-hospital decision-support case: keep patient-level computation local, coordinate learning globally, and make every rollout traceable.



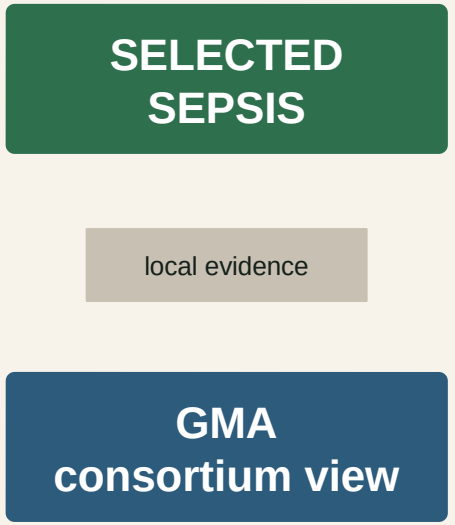
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Professor presentation

DOMAIN OPPORTUNITIES

Regulated domains need local execution and global coordination.

Healthcare and finance share fragmented data, strict access boundaries, drifting models, and audit-heavy operations. The proposed deep dive is multi-hospital sepsis decision support.

- Healthcare: sepsis/deterioration, readmission, chronic monitoring, claims quality, and imaging-model drift.
- Finance: payment fraud, AML pattern detection, credit-risk drift, liquidity, and transaction-operations anomalies.
- Why sepsis: bedside decisions must stay local while evidence and model governance improve across hospitals.
- Scope: retrospective and shadow-mode research first—no autonomous diagnosis or treatment.



WHY SEPSIS

Sepsis makes coordination a high-stakes problem.

The disease burden is large; the operational challenge is turning scattered, time-sensitive signals into governed clinical support.

48.9 million cases

Published 2020 global estimate summarized by WHO.

11 million deaths

WHO reports roughly 20% of global deaths in the cited estimate.

15 per 1,000

Estimated hospitalized patients developing sepsis as a complication of care.

Why it is hard

Signals span EHRs, labs, devices, and workflows that vary by hospital.

Evidence: WHO Sepsis fact sheet, 3 May 2024

THE CORE PROBLEM

Dagents makes the missing coordination layer explicit.

The problem is not only predicting sepsis; it is governing the path from local data to repeated multi-site action.

Fragmented signals

Different schemas, units, clocks, and quality rules. Dagents: SourceSpec plus LMA validation and profiling.

Site-specific reality

Patient mix, workflows, resources, and alert tolerance vary. Dagents: site-scoped LMAs plus GMA comparison.

Control-plane sprawl

Sites rebuild registration, scheduling, artifacts, deployment, and rollback. Dagents: shared contracts and services.

Evidence debt

Lineage, approvals, overrides, and audit evidence arrive late. Dagents: versioned plans, runs, artifacts, and rollback records.

Consequence: duplicated engineering, slower rollout, uneven behavior—and no reliable cross-site learning loop / 04

FUNCTIONAL SCOPE

The case needs a reusable system—not only a prediction model.

Five capability groups turn a candidate model into governed, multi-hospital clinical decision support.

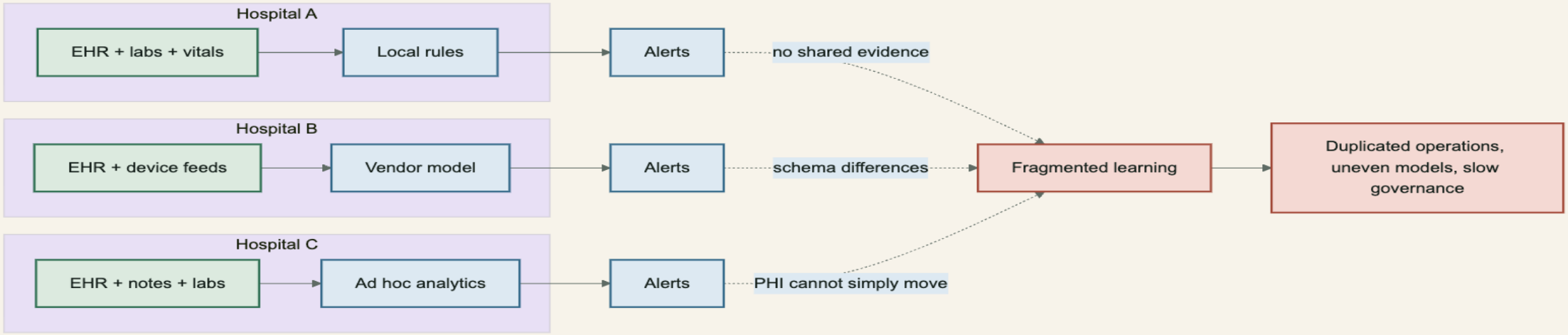
Ingest + quality	Local execution	Clinical workflow	Global coordination	Deploy + govern
FHIR/HL7 adapters, source validation, canonical features, provenance, units, freshness, and missingness.	LMA profiles approved data, scores locally, suppresses unsafe runs, and records patient-level traces.	Show recommendation basis, uncertainty, and limitations; capture acknowledgement, override, and outcome.	GMA registers sites, compares drift and calibration, and selects site-appropriate policies or bundles.	Pipelines, artifacts, typed plans, signed rollouts, canaries, monitoring, rollback, and audit evidence.

Cross-cutting controls: security, privacy, lineage, versioning, policy gates, retention, and clinician accountability.

CURRENT STATE

Hospitals solve the same risk problem in isolated stacks.

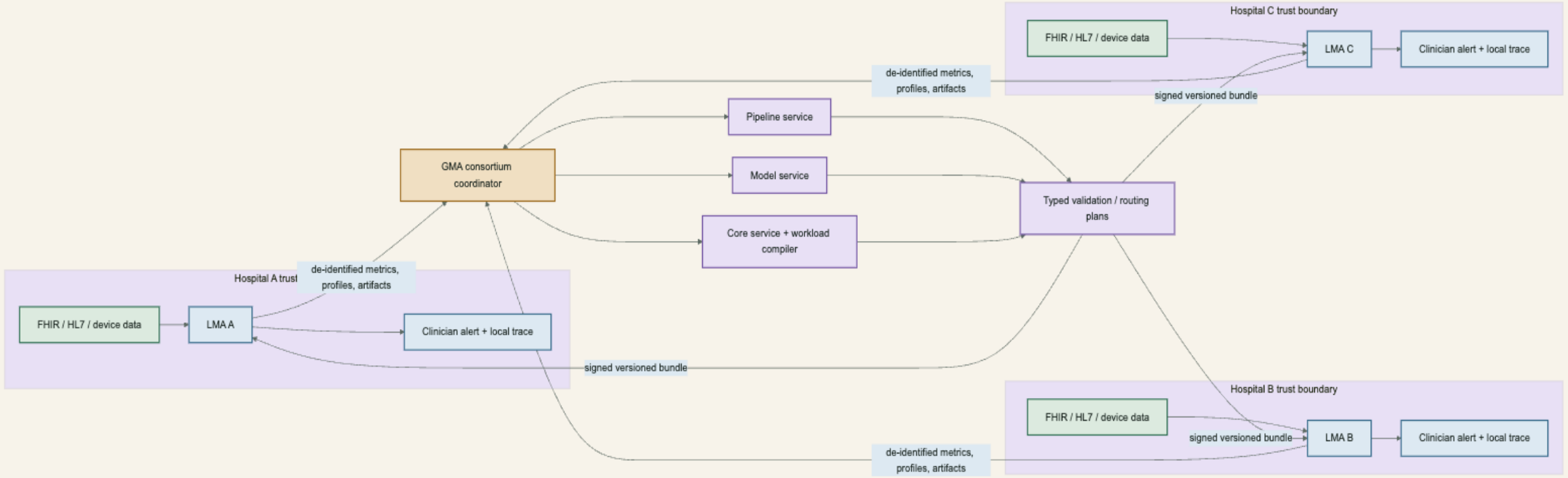
Site-specific rules and vendors limit shared learning; raw protected health information cannot simply be centralized.



TARGET ARCHITECTURE

Dagents scores locally and coordinates evidence globally.

LMAs run inside hospital trust boundaries; the GMA receives approved summaries and returns governed, versioned bundles.



LMA FLOW

The LMA owns the patient-level decision-support loop.

A local run validates scope and quality before scoring; the clinician retains decision authority.

INGEST

Receive approved FHIR, HL7, database, or event-stream data inside the hospital boundary.

VALIDATE

Check schema, units, provenance, freshness, cohort, consent or purpose scope, and missingness.

SCORE

Run the approved local model; return risk, uncertainty, model version, and input basis.

SUPPORT

Present evidence to a clinician; capture acknowledgement, deferral, override, action, and outcome.

Only approved de-identified metrics and artifact pointers leave the hospital trust boundary.

GMA FLOW

The GMA governs cross-hospital learning.

Aggregate coordination improves consistency without erasing hospital-specific populations, workflows, or risk tolerances.

REGISTER

Track each LMA identity, hospital scope, version, capability, and approved data-sharing policy.

ASSIMILATE

Accept approved profiles, performance metrics, run summaries, and artifact pointers with provenance.

COMPARE

Measure drift, calibration, subgroup behavior, alert burden, reliability, and configuration variance.

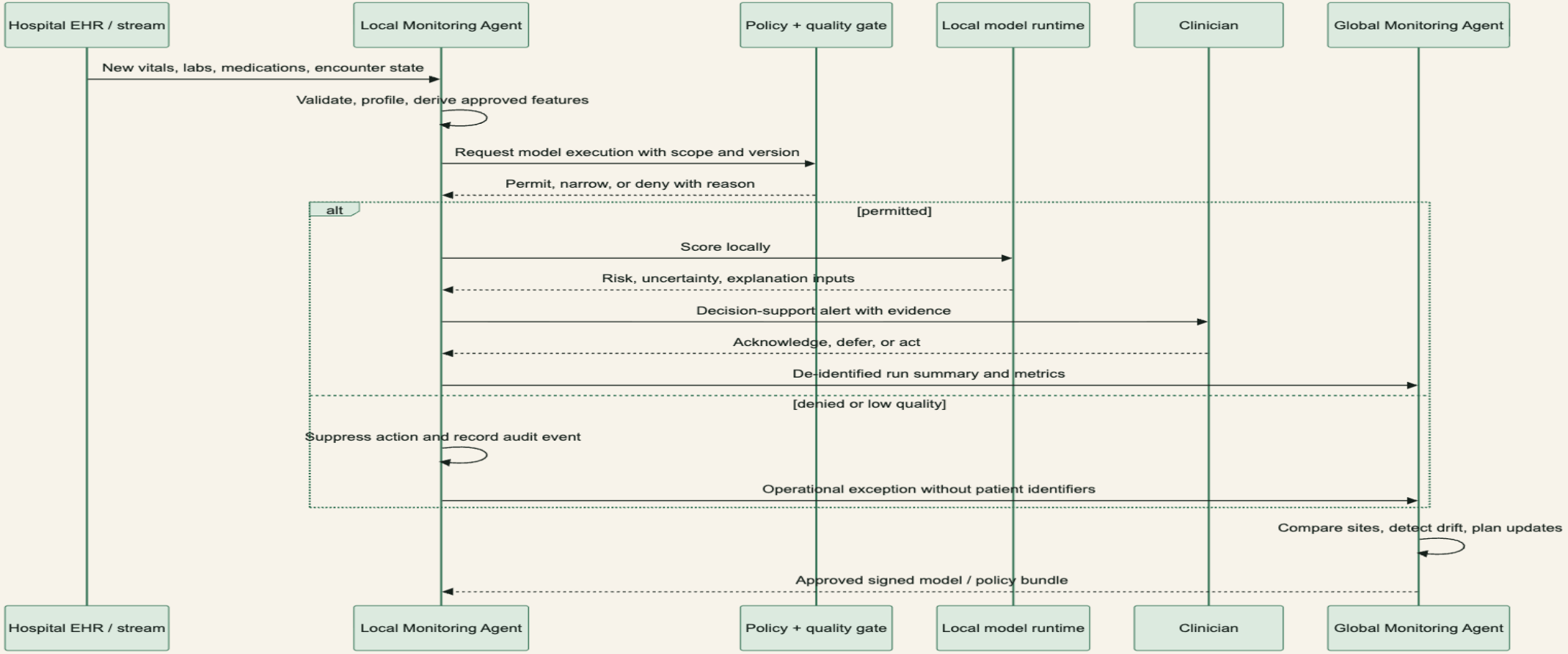
PLAN + ROLLOUT

Select site-appropriate bundles; compile plans, canary, monitor, pause, and roll back.

The GMA coordinates the fleet; it does not make bedside decisions or ingest unrestricted raw PHI.

A local decision becomes a global learning signal.

Policy and data-quality checks precede scoring; a clinician reviews the basis before any clinical workflow action.



CONTROL PLANE

Dagents already has the required control-plane shape.

Current contracts cover registration, profiling, model jobs, bundle synchronization, telemetry, and workload compilation.

REGISTER

GMA records hospital/LMA identity, scope, version, capability, and desired deployment state.

PROFILE

LMA profiles source-scoped records; GMA profiles approved assimilated evidence across sites.

RUN

Typed model-job requests identify task, model family, features, version, scope, and artifact prefix.

SYNC

Versioned bundles and plans coordinate desired state, deployment resources, health, and rollback.

Production work: durable state, broker-backed delivery, signed bundles, idempotency, and immutable audit evidence.

GOVERNANCE

Compliance requires shared controls.

Dagents can supply reusable technical enforcement and evidence; hospitals still own organizational, security, privacy, and clinical controls.

HIPAA

Dagents support: local PHI boundary, scoped sources, run IDs. Still needed: IAM/MFA, encryption and keys, retention, risk analysis, incident response, BAAs.

SOC 2

Dagents support: deterministic plans, health checks, change/run evidence. Still needed: control owners, access reviews, vendor risk, incident evidence, CPA examination.

CLINICAL AI

Dagents support: model/version trace, explanation inputs, policy gates, rollback. Still needed: intended use, clinical validation, human factors, regulatory and safety governance.

A production claim depends on implemented and operating controls—not architecture alone. SOC 2 is an attestation; HIPAA also includes administrative and physical safeguards.

RESPONSIBILITY MATRIX

Dagents enables evidence, not compliance.

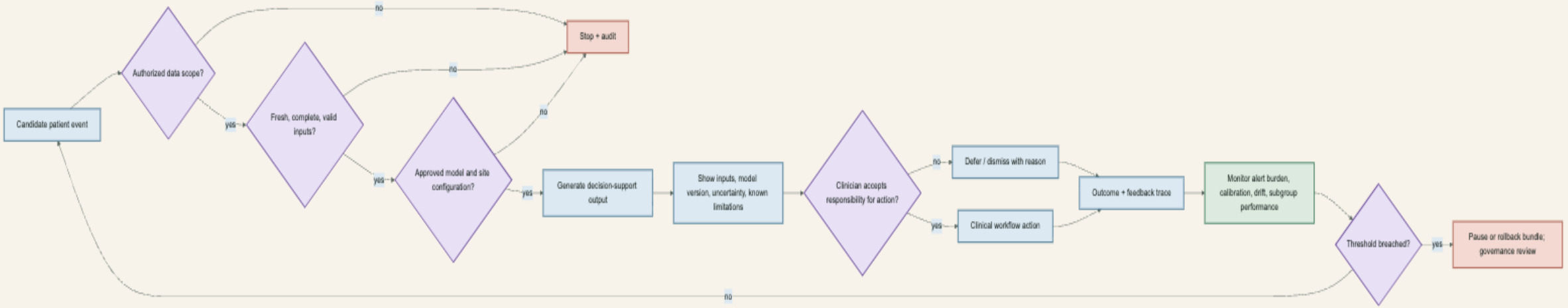
The implementation backlog should map each external obligation to a reusable Dagents mechanism and an accountable owner.

Framework	Dagents-aligned support	Production controls still required
HIPAA Security	Local boundary, scoped sources, versioned runs and plans	IAM/MFA, encryption and keys, audit controls, integrity, DR, incident response
HIPAA Privacy	Approved feature contracts; aggregate-only GMA interface	Purpose/minimum-necessary policy, consent or authorization, retention and deletion
SOC 2	Repeatable deployments, service health, evidence and lineage	Control ownership, access/change/vendor reviews, operating evidence, CPA report
Clinical CDS	Human decision point, model/input basis, validation metadata, rollback	Intended use, clinical validation, human factors, regulatory assessment
FHIR / USCDI	SourceSpec and canonical dataset boundaries fit adapter-based exchange	Profiles, terminology, conformance, provenance, patient matching

CLINICAL SAFETY

Clinical action requires four explicit gates.

Authorization, data quality, model approval, and human review are explicit; monitoring can pause or roll back a bundle.



Start retrospectively, then shadow mode. Fail closed, record the reason, keep the clinician accountable, and link monitoring thresholds to a tested pause/rollback plan.

EFFORT

Dagents moves effort toward clinical validation.

The framework adds discipline; the payoff grows with hospitals, models, deployments, and regulated consumers.

Concern	Reusable Dagents leverage	Case-specific or production work
Source boundary	Reuse SourceSpec, validation, profiling, and local/global scope contracts	Build FHIR/HL7 mappings, terminology, cohort, provenance, and patient matching
Orchestration	Reuse pipelines, runs, agent registration, telemetry, and workload plans	Add durable queues, retries, idempotency, replay, scheduling, and persistence
ML lifecycle	Reuse model-job, artifact, evaluation, and versioning patterns	Develop and clinically validate the sepsis model, thresholds, and monitoring
Deployment	Reuse Docker/Kubernetes compilation and desired-state concepts	Harden secrets, signing, isolation, canaries, rollback, and disaster recovery
Assurance	Reuse typed plans, run IDs, policy gates, and lineage points	Implement IAM, immutable logs, policies, training, audits, and clinical governance

For one static model at one site, Dagents may be more infrastructure than necessary / 15

DECISION GUIDE

Where Dagents fits—and where other tools lead.

Choose it when one control plane must span multiple data boundaries, model families, and deployment targets.

Option	Why teams choose it	Limit / Dagents fit
Dagents	LMA/GMA boundary plus pipelines, model jobs, typed plans, and workload compilation	Broad reusable control plane; less mature; hardening and clinical adapters required
Federated-learning toolkits	Flower / NVIDIA FLARE: distributed training, evaluation, privacy, and federation mechanics	Deeper FL, narrower lifecycle; use inside Dagents when collaborative training is needed
Managed MLOps SaaS	SageMaker AI / Azure ML: managed training, registries, endpoints, and monitoring	Fast cloud path; cloud coupling and hospital-boundary integration remain
Workflow platforms	Airflow / Kubeflow: DAGs, retries, caching, and Kubernetes execution	Mature workflows; no native LMA/GMA scope, capability registry, or clinical governance contract
Clinical point SaaS	Fastest when a validated product already fits the exact workflow	Narrow product/vendor boundary; Dagents fits multi-model, multi-site ownership

Dagents can complement these tools: it is the domain control plane around specialized runtimes / 16

PILOT ROADMAP

A safe pilot proves value in five controlled phases.

Evidence gates—not calendar dates—determine when the work advances from contracts to clinical use.

Phase	Goal	Evidence required
1. Define	Intended use, cohort, outcome label, exclusions, sources, and governance owners	Approved case specification and canonical data contract
2. Validate	Run retrospective LMAs separately at each hospital	Data-quality, calibration, subgroup, and baseline alert evidence
3. Shadow	Execute prospectively without showing recommendations to clinicians	Latency, reliability, alert volume, trace completeness, recovery results
4. Pilot	Show evidence to clinicians under an approved protocol	Workflow fit, burden, overrides, human factors, safety review
5. Scale	Let GMA compare sites; canary signed bundles and test rollback	Consortium results plus security, privacy, regulatory, and audit readiness

EVALUATION

Judge the pilot on four kinds of evidence.

Clinical performance alone is insufficient; safety, operations, and governance determine whether the system is usable and controllable.

Clinical usefulness

Discrimination and calibration; sensitivity and PPV at a fixed alert budget; lead time.

Safety + equity

Subgroup performance, alert burden, duplicates, acknowledgement, and override reasons.

Operational reliability

Freshness, missingness, run success, inference latency, recovery, and rollback time.

Governance evidence

Trace completeness, model/version coverage, access exceptions, and site-to-site variance.

Targets follow measured baselines; this deck does not invent clinical outcomes / 18

RECOMMENDATION

Use Dagents as the execution plane for a shadow-mode sepsis pilot.

The project is valuable because it separates local patient computation from global coordination—and makes every plan, model, run, and rollout traceable.

- LMAs keep patient-level processing close to hospital-controlled data.
- GMA coordinates approved evidence without becoming a central raw-PHI lake.
- Pipeline, model, core, and typed planner boundaries reduce duplicated platform work.
- HIPAA and SOC 2 readiness becomes a concrete control backlog with reusable evidence points.
- Clinical validation, security operations, regulatory analysis, and human oversight remain essential.